MILITARY SPECIFICATION

PLASTIC SHEET, VINYL, FLEXIBLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of colorless flexible vinyl sheet (see 6.1).

1.2 Classification.

Type I. - Thickness of 0.006 ± 0.0006 inches.
Type II. - Thickness of 0.012 ± 0.001 inches.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

STANDARDS

FEDERAL

FED-STD-123 - Marking for Domestic Shipping (Civil Agencies).
FED-STD-191 - Textile Test Methods.
FED-STD-406 - Plastics: Methods of Testing.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The vinyl sheet shall be nontoxic, shall be uniformly colorless and clear and shall not have an objectionable odor, and unless otherwise specified shall have a smooth, dull matte finish on both sides (see 6.2). The sheet shall be suitable compounded from virgin vinyl chloride resin and phosphate and phthalate plasticizers. Slight pigmentation will be permissible to neutralize any natural yellow tint. The use of water-soluble compounding ingredients is prohibited (see 6.3).

3.2 Thickness. The sheet shall be 0.006 ± 0.0006 or 0.012 ± 0.001 inches thick (Type I or Type II) when tested as specified in 4.4.4.2.

3.3 Bondability. The sheet shall be capable of being bonded to itself by commercially common, heat-or electronic-sealing devices and methods. The strength of such bonds at the seal and seal edge, in either the machine or transverse direction of the sheet, shall be no less than 3.0 pounds for Type I and no less than 13.0 pounds for Type II when tested as specified in 4.4.4.3. Bonded specimens shall be tested no less than 14 days after the date of bonding.

3.4 Dimensions.

3.4.1 Rolls. The width shall be as specified in the contract or order (see 6.2). A plus tolerance of 1/2 inch with no minus tolerance will be permissible. The width shall be measured on the manufacturer's rolls. The sheet shall be furnished in continuous lengths of no less than 50 yards.

3.4.2 Flat cuts. The width and length or other applicable dimensions of cuts, including tolerances, shall be as specified in the contract or order (see 6.2).

3.5 Physical properties. The sheet shall meet the physical requirements of table I.

3.6 Preproduction. Prior to the start of regular production, a preproduction sample of plastic sheet shall be produced, in accordance with this specification, for examination and tests (see 4.3).

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3.7 Workmanship. The sheet shall be clean, well finished, free from dirt, oil, foreign matter, rough or sharp edges, scratches, scuffs, cracks, creases, blisters, bubbles, pits, tears, cuts, holes, or other defects which may affect its appearance or serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise sspecified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1, 3.4 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of plastic sheet shall be produced, using the same process, formulation, and equipment as will be used during regular production.

4.3.2 Inspection procedure.

4.3.2.1 For examination. The preproduction sample shall be examined for all requirements of this specification.

4.3.2.2 For tests. The preproduction sample shall be tested in accordance with 4.4.4.

4.3.3 Tests. Test shall be conducted as specified in 4.4.4.

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Table I. Physical Requirements

Property	Unit	Requirements Type I	Type II	Test Paragraph
Tensile strength, each direction (minimum)	PSI	1600	2300	4.4.4.3
Ultimate elongation, machine direction (minimum)	PCT	175	200	4.4.4.3
Ultimate elongation, transverse direction (minimum)	PCT	200	250	4.4.4.3
Tear resistance, Elmendorf, each direction (minimum)	GRAMS	600	2200	4.4.4.4
Pinholes (maximum)	----	0	0	4.4.4.5
Extraction in soapy water (maximum)	PCT	10.0	10.0	4.4.4.6
Blocking at 158°F (maximum)	Scale rating	No. 3	No. 2	4.4.4.7
Volatility (maximum)	PCT	7.5	3.0	4.4.4.8
Resistance to weathering	No stiffening, cracking, discoloration, tackiness or exudation of plasticizer from the sheet.			4.4.4.9
Cold crack at -27° ± 2°F	No evidence of cracking			4.4.4.10
Lacquer lifting	No sign of lifting, tackiness, swelling in the area of contact, or exudation of plasticizer from the sheet.			4.4.4.11
Dimensional stability	No more than seven percent change in machine or transverse direction			4.4.4.12

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4.3.4 Acceptance/rejection criteria. The preproduction sample shall meet the examination and tests specified by 4.3.2 to be acceptable. The supplier shall obtain written approval from the contacting officer before proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the sheet produced in no more than three consecutive work shifts of one week by one manufacturer from the same materials and under essentially the same manufacturing conditions.

4.4.2 Sampling.

4.4.2.1 For examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105. The unit of product shall be one sheet or one linear yard.

4.4.2.2 For destructive tests.

4.4.2.2.1 Rolls. Sampling shall be conducted in accordance with MIL-STD-105, level S-1. Twenty-one square feet of material shall be taken from the end of each sample roll.

4.4.2.2.2 Flat cuts. Sampling shall be conducted in accordance with MIL-STD-105, level S-1.

4.4.3 Inspection procedure.

4.4.3.1 For examination and nondestructive tests. Sample sheet shall be examined in accordance with the classification of defects and MIL-STD-105.

4.4.3.2 For destructive tests. Sample sheet shall be tested in accordance with 4.4.4.3 and 4.4.4.4. The acceptance number is 0.

4.4.3.3 Classification of defects.

(a) Plastic Sheet, Vinyl, Flexible.

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 10 defects per hundred units	
101	Thickness incorrect	4.4.4.2

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4.4.3.3 Classification of defects (continued).

(a) Plastic Sheet, Vinyl, Flexible.

Categories	Defects	Acceptance standards
Major:	AQL 10 defects per hundred units	
102	Workmanship (see 3.7)	
	AQL 0.1 defect per hundred units	
103	Pinholes	4.4.4.5

4.4.4 Tests.

4.4.4.1 Specimens. No specimens shall be taken closer than 10 percent of the width of sheet from either edge. Five specimens shall be tested unless otherwise specified in the test method. Unless otherwise specified, test specimens shall be conditioned at a temperature of 73.5° ± 2°F and a relative humidity of 50 ± 4 percent for a period of no less than 16 hours.

4.4.4.2 Thickness. The thickness shall be determined using a dead weight type gage equipped with a dial graduated to read directly to 0.0001 inch. The presser foot shall be circular with a diameter of 0.35 ± 0.01 inch. The presser foot and moving parts connected therewith shall be weighted so as to apply a total load of 3 ± 0.1 ounce to the specimen. The presser foot and anvil surface shall be plane to within 0.0001 inch and parallel to each other within 0.0001 inch.

4.4.4.3 Bondability, tensile strength, and ultimate elongation. Bondability, tensile strength, and ultimate elongation shall be determined as specified in method 1013, method B of FED-STD-406 using a pendulum (Scott) type of machine, except that no autographic stress strain curve is needed. The test specimens shall be 1 inch wide. For breaking (bonding) strength of bonded seam, the seam shall be at the center of the specimen and perpendicular to the center line. The specimen shall be prepared for peel stress (bonding strength) by placing one strip of sheet (centered) on top of another and sealing one end, leaving the free ends to be gripped in the jaws of the testing machine.

4.4.4.4 Tear resistance, Elmendorf. Elmendorf tear resistance shall be determined as specified in method 5132 of FED-STD-191.

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4.4.4.5 Pinholes and cracks. Examination for pinholes and cracks shall be made by viewing the surface of the sheet held under light tension. A bright light source shall be positioned behind the sheet so as to clearly illuminate the surface without producing glare in the observer's eyes. A pinhole is defined as any opening observed in the sheet under the conditions specified above which is not visible when the sheet is viewed normally in average daylight or the equivalent thereof.

4.4.4.6 Extraction in soapy water. The extraction test shall be performed by immersing a weighed sample of sheet, 4 inches square, in 400 cubic centimeters of a 1 percent soap solution for 24 hours at 122° ± 2°F and determining weight loss. The sample shall have been previously conditioned for 3 hours at 122° ± 2°F, cooled to room temperature in a dessicator, weighed immediately upon removal and then placed in the test solution. Individual samples shall be tested in covered, one-pint glass containers. The samples must be held in such a manner that the entire surface is exposed to the test solution. Upon removal after the test period, the sample shall be gently wiped with a soft cloth or tissue paper. It shall then be reconditioned and weighed in the same manner as was done initially. The soap used shall be a neutral toilet soap. All weighings shall be made to the nearest 0.0005 gram.

4.4.4.7 Blocking. Blocking shall be determined as specified in method 5872 of FED-STD-191, except that the test shall be performed at 158° ± 2°F for a period of 48 hours.

4.4.4.8 Volatility. The volatility tests shall be performed according to method 6081 in FED-STD-406.

4.4.4.9 Resssistance to weathering. A sample of sheet shall be exposed as specified in method 5670 of FED-STD-191 for a period of 100 hours.

4.4.4.10 Cold crack. The cold crack tests shall be performed at the required temperatures as specified in method 5874 of FED-STD-191 after a conditioning period of not less than 1 hour at the test temperatures. After the test, the samples shall be examined visually at room temperature for cracking as specified in 4.4.4.5.

4.4.4.11 Lacquer lifting. The lacquer lifting test shall be conducted by placing a 3-inch by 5-inch sample of the vinyl sheet over a lacquered panel of equal or larger dimensions. The sheet shall be covered with a

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flat glass plate which is then weighted with a 2-pound weight. The sheet shall remain exposed for 14 days. The lacquered panels shall be prepared by supplying two medium coats of the following lacquer to thin carbon sheet steel, the surface of which is smooth and properly perpared to insure freedom from grease and foreign material:

Lacquer

	Parts by weight
Nitrocellulose, 1/2 sec.	16.0
Dibutyl phthalate	4.0
Ethyl alcohol	10.6
Toluene	40.0
Butyl acetate	16.0
Ethyl acetate	8.6
Butyl alcohol	4.8

The panels shall be air-dried for a minimum of 2 days prior to use.

4.4.4.12 Dimensional stability. Dimensional stability shall be determined by exposing a 10-inch square sample of sheet to 212° ± 2°F. for 30 ± 1 minutes in a circulating air oven. The sample shall be cut from the center of the sheet with a template accurate to ± 0.02 inches in all directions. The sample shall be placed between sheets of heavy wrapping paper, 15 inches square, lightly dusted with talc. The paper sheets shall be clipped or stapled together, care being taken not to hold or restrict the test sample. The sandwich shall then be placed horizontally in an oven. After the oven exposure, the sample shall be cooled to room temperature and measured to the nearest 0.01 inch along both central axis.

5. PREPARATION FOR DELIVERY

5.1 Interplant Shipment (see 6.4). The plastic sheet shall be packaged and packed to prevent damage in shipment from the supply source to the first receiving activity for immediate use or further processing. Shipping containers shall be in compliance with the rules and regulations applicable to the mode of transportation.

5.2 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with FED-STD-123, and shall include the contract or purchase order number, and name and address of the prime contractor.

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6. NOTES

6.1 Intended use. This vinyl sheet is intended for use as the bag material of the M1 and M1A1 Waterproofing Bag.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type I or Type II sheet.
(c) Whether rolls or flat cuts and sizes of each.
(d) Finish, if other than a smooth dull matte finish (see 3.1).
(e) Preproduction.

(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the supplier whether or not to proceed with production.

6.3 Ingredients. The compounding limitations are intended to insure adequate resistance of the sheet to mildew and bacterial degradation (see 3.1).

6.4 Interplant Shipment. Packaging and packing for supplies and materials that do not enter the military supply system. Typical interplant shipments are shipments from a vendor to a prime contractor, a subcontractor or a supplier to a military arsenal or plant.

6.5 Supersession data. This specification supersedes Edgewood Arsenal Purchase Description 197-54-839A, dated 5 April 1968.

Custodian:

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Preparing Activity:

Army - EA

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